1. ECG-RESTING

I.TEST PROCEDURE

Position : Supine
Recording : Record ECG at rest state
Condition : Subject should be in rest state
Duration : 60 Seconds

II.PARAMETER CALCULATION

Heart Rate: BPM

III.GRAPHICAL REPRESENTATION

Tachogram: RR_Interval (seconds) vs Time (seconds)

Histogram: RR_Interval (seconds) vs Count

Heart Rate (BPM) vs Count

2. ECG-DEEP BREATHING

I.TEST PROCEDURE

Position : Supine
Recording : Record ECG in Deep Breathing.
Condition : 5 Sec in & 5 Sec out 6 breaths per min
Duration : 60 Seconds

II.PARAMETER CALCULATION

Mean

Mean RR_Interval: Average of RR_Interval

Formula:

Mean RR_Interval = (Total of RR_Interval/No. of RR_Interval)

Mean Heart Rate: Average of Heart Rate

Formula:

Mean Heart Rate = (Total of Heart Rate /No. of Heart Rate)

Standard Deviation

SD of RR_Interval

Formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

σ = Standard Deviation of RR_Interval
X_i = RR_Interval
$\overline{X}$ = Mean RR_Interval
n = No. of RR_Interval

SD of Heart Rate:

Formula:

σ = Standard Deviation of Heart Rate
X_i = Heart Rate
$\overline{X}$ = Mean Heart Rate
n = No. of Heart Rate

<u>Coefficient of Variation</u>
Formula:

CV = {(SD of RR_Interval/Mean RR_Interval) x 100}

<u>Average Deep Breathing Difference (DBD)</u>

Formula:

DBD = Mean of (Max RR_Interval – Min RR_Interval in each respiratory period)

<u>Respiratory Sinus Arrhythmia (RSA)</u>

Formula:

RSA Index = ({DBD/ (RR1+RR2)}/2)

RR1- Mean of longest RR interval during expiration counted from 6 consecutive cycles

RR2- Mean of shortest RR interval during inspiration counted from 6 consecutive cycles

<u>Expiration-Inspiration Ratio (E: I)</u>

Formula:

E: I = (Mean of Longest RR_Intervals during Exhalation/Mean of Lowest RR_intervals during Inhalation)

<u>III.GRAPHICAL REPRESENTATION</u>

Tachogram: RR_Interval (seconds) vs Time (seconds)

Histogram: RR_Interval (seconds) vs Count

Heart Rate (BPM) vs Count

3. ECG-STANDING

I.TEST PROCEDURE

Position : Supine & Standing
Recording : Record ECG in Supine and Immediate Standing.
Condition : Immediate supine to standing up. In normal subject
Duration : 60 Seconds

II.PARAMETER CALCULATION

Mean

Mean RR_Interval: Average of RR_Interval

Formula:

Mean RR_Interval = (Total of RR_Interval/No. of RR_Interval)

Mean Heart Rate: Average of Heart Rate

Formula:

Mean Heart Rate = (Total of Heart Rate /No. of Heart Rate)

Standard Deviation

SD of RR_Interval

Formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

σ = Standard Deviation of RR_Interval
X_i = RR_Interval
$\overline{X}$ = Mean RR_Interval
n = No. of RR_Interval

SD of Heart Rate:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

Formula:

σ = Standard Deviation of Heart Rate
X_i = Heart Rate
$\overline{X}$ = Mean Heart Rate
n = No. of Heart Rate

<u>Coefficient of Variation</u>

Formula:

CV = {(SD of RR_Interval/Mean RR_Interval) x 100}

<u>30th Beat R-R Interval</u>

Formula:

30th beat RR_Interval = Longest RR_Interval around 30th beat at standing.

<u>15th Beat R-R Interval</u>

Formula:

15th beat RR_Interval = Shortest RR_Interval around 15th beat at standing.

<u>30:15 Ratio</u>

Formula:

30 : 15 Ratio = (Longest RR_Interval at 30th beat/ Shortest RR_Interval at 15th beat)

<u>III.GRAPHICAL REPRESENTATION</u>

Tachogram: RR_Interval (seconds) vs Time (seconds)

Histogram: RR_Interval (seconds) vs Count

Heart Rate (BPM) vs Time Count

4. ECG-VALSALVA MANEUVER

TEST PROCEDURE

Position : Sitting
Recording : Record ECG in Sitting with pressure gauge at 40mmHg.
Condition : Subject to sit quietly & then blow into a mouthpiece attached to pressure Gauge at a pressure 40mmHg for 15 Sec.
Duration : 60 Seconds

II.PARAMETER CALCULATION

Mean

Mean RR_Interval: Average of RR_Interval

Formula:

Mean RR_Interval = (Total of RR_Interval/No. of RR_Interval)

Mean Heart Rate: Average of Heart Rate

Formula:

Mean Heart Rate = (Total of Heart Rate /No. of Heart Rate)

Standard Deviation

SD of RR_Interval

Formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

σ = Standard Deviation of RR_Interval
X_i = RR_Interval
$\overline{X}$ = Mean RR_Interval
n = No. of RR_Interval

SD of Heart Rate:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

Formula:

σ = Standard Deviation of Heart Rate
X_i = Heart Rate
$\overline{X}$ = Mean Heart Rate
n = No. of Heart Rate

<u>Coefficient of Variation</u>

Formula:

CV = {(SD of RR_Interval/Mean RR_Interval) x 100}

<u>Max R-R Interval</u>

Formula:

Max RR_Interval = Longest RR_Interval after the Maneuver.

<u>MIN R-R Interval</u>

Formula:

Min RR_Interval = Shortest RR_Interval during the maneuver.

<u>VALSALVA RATIO</u>

Formula:

Max RR_Interval = (Longest RR_Interval after the Maneuver / Shortest RR_Interval during the maneuver).

<u>III.GRAPHICAL REPRESENTATION</u>

Tachogram: RR_Interval (seconds) vs Time (seconds)

Histogram: RR_Interval (seconds) vs Count

Heart Rate (BPM) vs Count

5.POSTURAL HYPOTENSION

I.TEST PROCEDURE

Position : Supine & Standing
Recording : Monitor BP in Supine and immediate Standing and after 60 seconds
Condition : Immediate supine to standing up.

II.PARAMETER CALCULATION

BP Supine : mmHg

BP Standing Immediate: mmHg

BP After 60 Seconds : mmHg

Difference in Systolic : mmHg

6. SUSTAINED HANDGRIP

I.TEST PROCEDURE

Position : Sitting
Recording : Monitor BP in Sitting in dominant arm and Immediate BP recording with Holding the bar.
Condition : Subject should hold the bar in their non-dominant arm as tightly as possible.

II.PARAMETER CALCULATION

BP Before Handgrip : mmHg

BP After Handgrip : mmHg

Difference in Diastolic : mmHg

INTERPRETATION VALUES

I.PARASYMPATHETIC FUNCTION

1. RESTING

RR_Interval

Normal Range: (0.6 Sec ≤ RR_Interval ≥ 1 Sec)
Abnormal Range: (< 0.6 Sec & > 1 Sec)

Heart Rate

Normal Range: (60 BPM ≤ Heart Rate ≥ 100 BPM)
Abnormal Range: (< 60 BPM & > 100 BPM)

2. DEEP BREATHING

Interpretation: Heart Rate response to deep breathing (MaxHR-MinHR)

Normal Range: >15 BPM
Borderline: 11 BPM ≤ Heart Rate ≤ 14 BPM
Abnormal Range: < 10 BPM

3. RESPONSE TO STANDING

Interpretation: Heart Rate response to standing up (30:15Ratio)

Normal Range: >1.04 BPM
Borderline: 1.01 BPM ≤ Heart Rate ≤ 1.03 BPM
Abnormal Range: < 1.00 BPM

4. VALSALVA MANEUVER

Interpretation: Heart Rate response to strain

Normal Range: > 1.21 BPM
Borderline: Nil
Abnormal Range: < 1.20 BPM

II.SYMPATHETIC FUNCTION

1. POSTURAL HYPOTENSION

Interpretation: BP response to standing up (fall in systolic BP)

Normal Range: < 10 mmHg
Borderline: 11 mmHg ≤ BP ≤ 29 mmHg
Abnormal Range: > 30 mmHg

2. SUSTAINED HANDGRIP

Interpretation: BP response to sustained handgrip (rise in diastolic BP)

Normal Range: > 16 mmHg
Borderline: 11 mmHg ≤ BP ≤ 15 mmHg
Abnormal Range: < 10 mmHg

HRV ANALYSIS

To capture 5 Minutes ECG and to calculate the following parameters

Time Domain Results

1. Mean

Mean RR_Interval: Average of RR_Interval

Formula:

Mean RR_Interval = (Total of RR_Interval/No. of RR_Interval)

Mean Heart Rate: Average of Heart Rate

Formula:

Mean Heart Rate = (Total of Heart Rate /No. of Heart Rate)

2. Standard Deviation

SD of RR_Interval

Formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

σ = Standard Deviation of RR_Interval
X_i = RR_Interval
$\overline{X}$ = Mean RR_Interval
n = No. of RR_Interval

3. SD of Heart Rate:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}}$$

Formula:

σ = Standard Deviation of Heart Rate
X_i = Heart Rate
$\bar{X}$ = Mean Heart Rate
n = No. of Heart Rate

4. Coefficient of Variation

Formula:

CV = {(SD of RR_Interval/Mean RR_Interval) x 100}

5. RMSSD

Root mean square of successive differences (square root of the mean squared differences of successive RR intervals)

$$RMSSD = \sqrt{\frac{1}{N-1}\sum_{j=1}^{N-1}(RR_{j+1} - RR_j)^2}$$

6. NN50

Count of adjacent RR intervals differing by more than 50 ms in the entire analysis interval.

$$NN50 = \sum_{i=1}^{N}\{|RR_{i+1} - RR_i| > 50ms\}[count]$$

Where N is total number of all RR intervals in segment

7. PNN50

NN50 count divided by total number of all RR intervals in selected segment

$$pNN50 = \frac{NN50}{N} \cdot 100[\%]$$

Where N is total number of all RR intervals in segment

Frequency Domain Results

1. FFT Spectrum (Welch's periodogram: 256 s window with 50% overlap)
2. VLF (0-0.04Hz)- Very low frequency band
3. LF (0.04-0.15Hz)- Low frequency band
4. HF (0.15-0.4Hz)- High frequency band
5. Total Power- Total power of the spectrum
6. LF/HF ratio
7. Normalized LF & Normalized HF

HF norm= HF/ (TP-VLF)*100= HF/ (LF+HF)*100

LF norm= LF/ (TP-VLF)*100= LF/ (LF+HF)*100